

# Olympiad Test Paper — Mathematics (Class 7) — Paper 4

## Chapter 4: Expressions Using Letter-Numbers

### Paper 4 — (progressively harder)

|                        |                                      |
|------------------------|--------------------------------------|
| <b>Subject</b>         | Mathematics (NCERT Class 7)          |
| <b>Chapter</b>         | 4 — Expressions Using Letter-Numbers |
| <b>Total Questions</b> | 60 (Multiple Choice, single correct) |
| <b>Maximum Marks</b>   | 240                                  |
| <b>Suggested Time</b>  | 90 minutes                           |

**Marking scheme:** +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

**Instructions:** Each question has four options (A), (B), (C), (D); exactly one is correct. Watch order of operations under awkward substitutions, never merge unlike terms or drop a constant, distribute a coefficient over every inside term (including subtraction signs), and generalise a pattern from its actual differences rather than guessing. Do not write on this paper — mark answers on your answer sheet.

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**1.** A photo studio charges a sitting fee of 90 rupees that already includes the very first print, then 24 rupees for every print after the first. If a customer orders  $p$  prints ( $p \geq 1$ ), the total charge in rupees is:

- (A)  $24p + 90$  (B)  $90(p - 1) + 24$   
(C)  $24(p - 1) + 90$  (D)  $114p$

**2.** Evaluate  $2a^2 - ab - b^2$  when  $a = -3$  and  $b = 2$ .

- (A) 20 (B) 8  
(C) 32 (D) -4

**3.** Simplify  $3(2x - 5y) - 2(4x - 3y) + y$ .

- (A)  $-2x - 14y$  (B)  $-2x - 8y$   
(C)  $2x - 8y$  (D)  $-2x + y$

**4.** Which expression is EQUIVALENT to  $18 - 12m$ ?

- (A)  $6(3 - 12m)$  (B)  $6(3 - 2m)$   
(C)  $-6(2m + 3)$  (D)  $12(m - 18)$

**5.** A pattern of beads: figure 1 has 2 beads, figure 2 has 5, figure 3 has 8, figure 4 has 11 — each figure adds 3. The rule for the  $n$ th figure is:

(A)  $3n - 1$

(B)  $3n$

(C)  $3n + 2$

(D)  $n + 3$

**6.** A library fines a reader 5 rupees on the first late day and 3 rupees for each later late day. A reader is late by  $d$  days ( $d \geq 1$ ). The total fine in rupees is:

(A)  $3d + 5$

(B)  $5d + 3$

(C)  $3(d - 1) + 5$

(D)  $5(d - 1) + 3$

**7.** Evaluate  $(p - q)^2 - 2pq$  when  $p = 5$  and  $q = -2$ .

(A) 69

(B) 29

(C) 49

(D) 9

**8.** Simplify  $5a - \{3a - [2a - (a - 4)]\}$ .

(A)  $3a - 4$

(B)  $5a - 4$

(C)  $3a + 4$

(D)  $a - 4$

**9.** An auditorium's rows each have 2 more seats than the row in front, and the front row holds 18 seats. The number of seats in the row that is 4 rows behind the front is:

(A)  $18 + 4 = 22$

(B)  $18 - 8 = 10$

(C)  $18 \times 4 = 72$

(D)  $18 + 8 = 26$

**10.** A sequence has  $n$ th term  $n^2 + 1$ : 2, 5, 10, 17, ... . The DIFFERENCE between the 7th term and the 5th term is:

(A) 24

(B) 12

(C) 26

(D) 4

**11.** A wholesaler sells rice in  $n$  large sacks at 800 rupees each and  $(n + 3)$  small sacks at 300 rupees each, then gives a flat 200-rupee loyalty discount on the whole order. The total cost in rupees, simplified, is:

(A)  $1100n + 900$

(B)  $1100n^2 + 700$

(C)  $1100n + 1100$

(D)  $1100n + 700$

**12.** Evaluate  $(3x - 2y) \div 2 + xy$  when  $x = 4$  and  $y = -1$ .

(A) 11

(B) 3

(C) 1

(D) 7

**13.** Simplify  $4(x + 2) - 3(2x - 1) + 2(x - 5)$ .

(A)  $-1$

(B)  $4x + 1$

(C) 1

(D)  $12x - 4$

**14.** Which expression is EQUIVALENT to  $2(3x - 4) + 3(x + 2)$ ?

(A)  $9x + 2$

(B)  $9x - 2$

(C)  $9x - 14$

(D)  $5x - 2$

**15.** A staircase pattern of blocks has stage totals 1, 3, 6, 10, 15, ... (the increases are 2, then 3, then 4, then 5). The number of blocks ADDED when going from stage 5 to stage 6 is:

(A) 6

(B) 5

(C) 15

(D) 21

**16.** Three pipes fill a tank: pipe A adds  $x$  litres, pipe B adds 10 litres more than twice A, and pipe C adds half of what A adds. The total litres added, simplified, is:

(A)  $3x + 10$

(B)  $(5/2)x + 10$

(C)  $(7/2)(x + 10)$

(D)  $(7/2)x + 10$

**17.** Evaluate  $a^2b - ab^2$  when  $a = 2$  and  $b = -3$ .

(A) 30

(B) -30

(C) 6

(D) -6

**18.** Simplify  $7p - 3q + 5 - (2p - 4q + 9) + q$ .

(A)  $5p + 2q - 4$

(B)  $5p - 8q + 14$

(C)  $5p + 2q + 14$

(D)  $9p + 2q - 4$

**19.** Which expression is NOT equivalent to  $24x - 36$ ?

(A)  $12(2x - 3)$

(B)  $4(6x - 9)$

(C)  $6(4x - 9)$

(D)  $24x - 36$

**20.** A pattern has stage dots 5, 8, 11, 14, ... increasing by 3. The SUM of the dots in stages 1, 2 and 3 together, and the  $n$ th-stage rule, are:

(A) 24 dots; rule  $3n$

(B) 24 dots; rule  $3n + 2$

(C) 21 dots; rule  $3n + 2$

(D) 24 dots; rule  $n + 3$

**21.** In the expression  $7 - 5x + 2x^2 - x^3$ , the coefficient of  $x$  and the constant term are, respectively:

(A) 5 and 7

(B) -5 and -1

(C) -5 and 7

(D) -1 and 7

**22.** Evaluate  $100 - 2n^2 + 3n$  when  $n = -4$ .

(A) 56

(B) 80

(C) 120

(D) 44

**23.** Simplify  $3x^2 + 5x - 2 - x^2 + 2x - 7$ .

(A)  $2x^2 + 7x - 9$

(B)  $2x^4 + 7x - 9$

(C)  $2x^2 + 7x + 5$

(D)  $4x^2 + 7x - 9$

**24.** A delivery rule charges 60 rupees for the first 2 kilometres as one block and 15 rupees for each kilometre beyond 2. For a trip of  $d$  kilometres ( $d \geq 2$ ), the fare in rupees is:

(A)  $15d + 60$

(B)  $60(d - 2) + 15$

(C)  $15(d - 2) + 60$

(D)  $75d - 30$

**25.** Which expression is EQUIVALENT to  $5(2a - 3) - 2(a + 4)$ ?

(A)  $8a - 7$

(B)  $8a - 23$

(C)  $8a - 11$

(D)  $12a - 23$

**26.** A square pattern of tiles: figure  $n$  is an  $n$ -by- $n$  square with its 4 corner tiles removed. The number of tiles in figure 5 is:

(A) 25

(B) 20

(C) 21

(D) 16

**27.** Evaluate  $(2a + b)^2 \div b$  when  $a = 1$  and  $b = 4$ .

(A) 18

(B) 9

(C) 16

(D) 8

**28.** Simplify  $4(a - 2b) - 2(2a - 3b) + 5b$ .

(A)  $3b$

(B)  $-3b$

(C)  $8a + 3b$

(D)  $-2b$

**29.** A theatre seats people in rows: row 1 has 12 seats, and each later row has 4 more seats than the row before. Using  $r$  for the row number, the number of seats in row  $r$  is:

(A)  $4r + 12$

(B)  $12r + 4$

(C)  $4r - 4$

(D)  $4r + 8$

**30.** Joining  $n$  unit squares in a single straight row gives a figure of perimeter  $2n + 2$  units. How much GREATER is the perimeter of the 10-square row than the 6-square row?

(A) 4

(B) 10

(C) 22

(D) 8

**31.** A canteen charges  $f$  rupees for a meal. A student buys a meal each weekday (5 days) and on 2 of those days also buys a 30-rupee dessert. Over one week the total spent, simplified, is:

(A)  $5f + 30$

(B)  $5f + 150$

(C)  $5f + 60$

(D)  $7f + 60$

**32.** Evaluate  $(x + y)/2 - (x - y)/2$  when  $x = 9$  and  $y = -5$ .

(A) 9

(B) 5

(C) 7

(D)  $-5$

**33.** Simplify  $(3a + 2b) - (a - b) - (2a + 4b)$ .

(A)  $-b$

(B)  $6a + 7b$

(C)  $-b + 6a$

(D)  $-7b$

**34.** Which expression correctly represents 'a number  $x$  is first tripled, then 4 is subtracted, and the whole result is doubled'?

(A)  $2(3x) - 4$

(B)  $2(3x - 4)$

(C)  $3(2x - 4)$

(D)  $6x - 4$

- 35.** A square of side  $s$  has border tiles  $4s - 4$ . A growing square frame is built so that figure  $n$  is the border of an  $(n + 2)$ -by- $(n + 2)$  square. For figure 3, the number of border tiles is:
- (A) 20 (B) 16  
(C) 25 (D) 12
- 36.** Evaluate  $3p^2 - 2pq + q^2$  when  $p = -2$  and  $q = 3$ .
- (A) 9 (B) 21  
(C) 33 (D) -3
- 37.** Simplify  $2(x + 3) - [4 - (x - 1)] + x$ .
- (A)  $4x + 9$  (B)  $2x + 1$   
(C)  $4x - 1$  (D)  $4x + 1$
- 38.** Which expression is EQUIVALENT to  $3(2x - 1) - (x - 5)$ ?
- (A)  $5x + 2$  (B)  $5x - 2$   
(C)  $5x - 8$  (D)  $7x + 2$
- 39.** A sequence has  $n$ th term  $2n + 1$ : 3, 5, 7, 9, ... The SUM of the first 4 terms is:
- (A) 20 (B) 16  
(C) 9 (D) 24
- 40.** A florist makes bouquets: each has  $r$  roses, twice as many daisies as roses, plus 3 ferns. The total number of flowers in  $m$  identical bouquets, simplified, is:
- (A)  $3mr + 3m$  (B)  $3r + 3$   
(C)  $2mr + 3m$  (D)  $3mr + 3$
- 41.** Evaluate  $5 - 2(3 - x)^2$  when  $x = 1$ .
- (A) 13 (B) -11  
(C) -3 (D) 9
- 42.** Simplify  $6m - 2n - (3m - 5n) + (m - n)$ .
- (A)  $4m - 8n$  (B)  $10m + 2n$   
(C)  $4m + 2n$  (D)  $4m + 4n$
- 43.** Which expression is EQUIVALENT to  $(1/2)(6x - 8) + (1/3)(9x + 6)$ ?
- (A)  $6x - 2$  (B)  $6x + 2$   
(C)  $6x + 14$  (D)  $3x - 2$
- 44.** A pattern has dot counts 1, 4, 9, 16, ... (square numbers). The number of dots ADDED going from figure  $n$  to figure  $(n + 1)$  is:
- (A)  $n + 1$  (B)  $2n + 1$   
(C)  $2n$  (D)  $n^2$
- 45.** A printing job costs a 250-rupee setup that includes the first 50 pages, then 4 rupees for each page beyond 50. For a job of  $t$  pages ( $t \geq 50$ ), the cost in rupees is:

- (A)  $4(t - 50) + 250$  (B)  $4t + 250$   
(C)  $4(t - 50) + 50$  (D)  $250t + 4$

**46.** Evaluate  $ab + bc + ca$  when  $a = 2$ ,  $b = -1$  and  $c = 3$ .

- (A)  $-1$  (B)  $7$   
(C)  $1$  (D)  $11$

**47.** Simplify  $5(2a + 3) - 4(a + 2) - (3a - 1)$ .

- (A)  $3a + 6$  (B)  $11a + 8$   
(C)  $3a + 8$  (D)  $3a + 10$

**48.** Four expressions are evaluated at  $x = 3$ :  $A = 2x^2 - 5$ ,  $B = 5x + 1$ ,  $C = x^3 - 10$ ,  $D = 4x + 4$ . Which has the GREATEST value?

- (A)  $A = 2x^2 - 5$  (value 13) (B)  $C = x^3 - 10$  (value 17)  
(C)  $B = 5x + 1$  (value 16) (D)  $D = 4x + 4$  (value 16)

**49.** A toymaker uses  $k$  kilograms of clay per doll and wastes a fixed 6 kilograms each batch. Making  $(k + 2)$  dolls in a batch uses, in kilograms, simplified:

- (A)  $k^2 + 6$  (B)  $3k + 6$   
(C)  $k^2 + 2k + 8$  (D)  $k^2 + 2k + 6$

**50.** If  $a = 4$ , the values of  $3a^2$ ,  $(3a)^2$  and  $3a + 3$  are, respectively:

- (A) 144, 144, 15 (B) 48, 144, 15  
(C) 48, 48, 15 (D) 48, 144, 27

**51.** Simplify  $(4x - y) - 2(x - 3y) + 3(y - x)$ .

- (A)  $-x + 8y$  (B)  $-x + 2y$   
(C)  $9x + 8y$  (D)  $-x - 4y$

**52.** A sequence has  $n$ th term  $4n - 1$ : 3, 7, 11, 15, ... . Which term number gives 39?

- (A) 9th (B) 39th  
(C) 11th (D) 10th

**53.** A market stall's takings: Friday  $6c + 2d$ , Saturday twice Friday's amount, Sunday  $4c - 3d$ . The simplified total in terms of  $c$  and  $d$  is:

- (A)  $16c + d$  (B)  $22c + 6d$   
(C)  $22c + 3d$  (D)  $22c - 3d$

**54.** Evaluate  $(2x - 1)(x + 3)$  by first finding each bracket then multiplying, using  $x = 2$ .

- (A) 10 (B) 3  
(C) 8 (D) 15

**55.** In the expression  $-x^2 + 6x - 9$ , which statement is correct?

- (A) The coefficient of  $x$  is  $-6$  and the constant term is 9  
(B) The coefficient of  $x^2$  is 1 and the constant term is 9

- (C) The coefficient of  $x$  is 6 and the constant term is 9  
(D) The coefficient of  $x$  is 6 and the constant term is  $-9$

**56.** Four expressions are evaluated at  $x = 4$ :  $P = x^2$ ,  $Q = 2x + 8$ ,  $R = 4x + 1$ ,  $S = 3x + 6$ . Exactly one pair below is equal in value for THIS substitution. Which pair is equal?

- (A) R and S (17 and 18)                      (B) P and R (16 and 17)  
(C) Q and S (16 and 18)                      (D) P and Q (both 16)

**57.** Simplify  $3xy - 2x + 4xy - 5x + xy - x$ .

- (A)  $8xy + 8x$                                   (B)  $8xy - 8x$   
(C)  $8x^2y - 8x$                                   (D)  $6xy - 8x$

**58.** Evaluate  $a^3 - b^3$  when  $a = 3$  and  $b = -2$ .

- (A) 19                                              (B) 23  
(C) 31                                              (D) 35

**59.** In a tower of bricks the  $n$ th layer from the top has  $(2n - 1)$  bricks (top layer 1, then 3, then 5, ...). The TOTAL number of bricks in the top 4 layers is:

- (A) 7                                                (B) 16  
(C) 9                                                (D) 20

**60.** Simplify  $7(a - 1) - 3(2a - 5) + 2(a + 1)$  and state its value when  $a = 0$ .

- (A)  $3a + 10$ , giving 0                      (B)  $3a - 10$ , giving  $-10$   
(C)  $3a + 22$ , giving 22                      (D)  $3a + 10$ , giving 10

# Solutions — Mathematics (Class 7), Chapter 4: Expressions Using Letter-Numbers — Paper 4

## Paper 4 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

## Answer Key

| Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans |
|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|
| 1  | C   | 11 | D   | 21 | C   | 31 | C   | 41 | C   | 51 | A   |
| 2  | A   | 12 | B   | 22 | A   | 32 | D   | 42 | C   | 52 | D   |
| 3  | B   | 13 | C   | 23 | A   | 33 | A   | 43 | A   | 53 | C   |
| 4  | B   | 14 | B   | 24 | C   | 34 | B   | 44 | B   | 54 | D   |
| 5  | A   | 15 | A   | 25 | B   | 35 | B   | 45 | A   | 55 | D   |
| 6  | C   | 16 | D   | 26 | C   | 36 | C   | 46 | C   | 56 | D   |
| 7  | A   | 17 | B   | 27 | B   | 37 | D   | 47 | C   | 57 | B   |
| 8  | C   | 18 | A   | 28 | A   | 38 | A   | 48 | B   | 58 | D   |
| 9  | D   | 19 | C   | 29 | D   | 39 | D   | 49 | D   | 59 | B   |
| 10 | A   | 20 | B   | 30 | D   | 40 | A   | 50 | B   | 60 | D   |

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

## Worked Solutions

**1. (C)** The 90 covers the first print, so only the remaining  $(p - 1)$  prints are charged at 24 each:  $24(p - 1) + 90$ . Treating all  $p$  prints as paid gives  $24p + 90$ , which double-charges the included first print.

**2. (A)**  $a^2 = 9$  so  $2a^2 = 18$ ;  $ab = (-3)(2) = -6$  so  $-ab = +6$ ;  $b^2 = 4$  so  $-b^2 = -4$ . Total  $18 + 6 - 4 = 20$ . Treating  $-ab$  as  $-6$  (forgetting the negative  $a$  flips the sign) gives  $18 - 6 - 4 = 8$ .

**3. (B)**  $3(2x - 5y) = 6x - 15y$ ;  $-2(4x - 3y) = -8x + 6y$ . Adding with the lone  $+y$ :  $(6x - 8x) + (-15y + 6y + y) = -2x - 8y$ . Forgetting to change the sign of  $-3y$  inside the second bracket gives  $-15y - 6y + y = -20y$ ; merging  $x$  and  $y$  gives other traps.

**4. (B)**  $6(3 - 2m) = 18 - 12m$  by distributing 6 over both terms. Test  $m = 1$ :  $18 - 12 = 6$  and  $6(3 - 2) = 6$  match.  $6(3 - 12m) = 18 - 72m$  distributes only the first term;  $-6(2m + 3) = -12m - 18$ .



**5. (A)** The common difference is 3, so the rule is  $3n + c$ . At  $n = 1$  the value is 2, and  $3(1) + c = 2$  gives  $c = -1$ , so  $3n - 1$ . Using just  $3n$  gives 3 at  $n = 1$  (one too many);  $3n + 2$  gives 5 at  $n = 1$ .

**6. (C)** Day 1 costs 5; the remaining  $(d - 1)$  days cost 3 each:  $3(d - 1) + 5$ . Test  $d = 1$ :  $3(0) + 5 = 5$  correct. The trap  $3d + 5$  gives 8 for a single late day.

**7. (A)**  $p - q = 5 - (-2) = 7$ , so  $(p - q)^2 = 49$ ;  $2pq = 2(5)(-2) = -20$ , so  $-2pq = +20$ . Total  $49 + 20 = 69$ . Mishandling the  $-2pq$  sign as  $-20$  gives  $49 - 20 = 29$ ; reading  $p - q$  as 3 gives 9-style errors.

**8. (C)** Innermost:  $2a - (a - 4) = 2a - a + 4 = a + 4$ . Next:  $3a - (a + 4) = 2a - 4$ . Outer:  $5a - (2a - 4) = 5a - 2a + 4 = 3a + 4$ . Test  $a = 0$ :  $5(0) - \{0 - [0 - (-4)]\} = -(-4) = 4 = 3(0) + 4$  correct.

**9. (D)** Each row back adds 2 seats; 4 rows back adds  $4 \times 2 = 8$  seats:  $18 + 8 = 26$ . Adding only 4 forgets the 2-per-row step; subtracting moves the wrong direction.

**10. (A)** 7th term =  $49 + 1 = 50$ ; 5th term =  $25 + 1 = 26$ ; difference  $50 - 26 = 24$ . The +1 cancels, leaving  $49 - 25 = 24$ . Using  $(7 - 5)^2 + \text{something}$  or just  $(7 - 5)$  gives small wrong values.

**11. (D)** Large:  $800n$ . Small:  $300(n + 3) = 300n + 900$ . Sum  $1100n + 900$ , then subtract the 200 discount:  $1100n + 900 - 200 = 1100n + 700$ . Forgetting the discount leaves  $1100n + 900$ .

**12. (B)**  $3x - 2y = 12 - (-2) = 14$ , so  $\div 2 = 7$ ;  $xy = 4(-1) = -4$ . Total  $7 + (-4) = 3$ . Reading  $-2y$  as  $-2$  gives  $10 \div 2 + (-4) = 1$ ; dropping the  $xy$  sign gives  $7 + 4 = 11$ .

**13. (C)**  $4(x + 2) = 4x + 8$ ;  $-3(2x - 1) = -6x + 3$ ;  $2(x - 5) = 2x - 10$ . x-terms  $4x - 6x + 2x = 0$ . Constants  $8 + 3 - 10 = 1$ . The x-terms cancel, leaving the constant 1. Test  $x = 5$ :  $28 - 27 + 0 = 1$  correct.

**14. (B)**  $2(3x - 4) = 6x - 8$ ;  $3(x + 2) = 3x + 6$ . Sum  $(6x + 3x) + (-8 + 6) = 9x - 2$ . Test  $x = 1$ :  $2(-1) + 3(3) = 7$  and  $9 - 2 = 7$  match. Dropping the +6 gives  $9x - 14$ .

**15. (A)** The increases are 2, 3, 4, 5, 6, ... (one more each stage). Stage 5 total is 15; the step into stage 6 is the next number, 6 (giving total 21). Using 5 repeats the previous step; 21 is the stage-6 total, not the increase.

**16. (D)**  $A = x$ ;  $B = 2x + 10$ ;  $C = x/2$ . Total  $x + 2x + 10 + x/2 = (1 + 2 + 1/2)x + 10 = (7/2)x + 10$ . Forgetting pipe C's half gives  $3x + 10$ ; forgetting pipe A's own  $x$  gives  $(5/2)x + 10$ .

**17. (B)**  $a^2b = 4(-3) = -12$ ;  $ab^2 = 2(9) = 18$ , so  $-ab^2 = -18$ . Total  $-12 - 18 = -30$ . Reading  $ab^2$  as  $2(-9) = -18$  (squaring then keeping the negative) gives  $-12 + 18 = 6$ .

- 18. (A)** Remove the bracket changing every inside sign:  $-2p + 4q - 9$ . Combine: p-terms  $7p - 2p = 5p$ ; q-terms  $-3q + 4q + q = 2q$ ; constants  $5 - 9 = -4$ . Result  $5p + 2q - 4$ . Not flipping the bracket signs gives  $5p - 8q + 14$ .
- 19. (C)**  $6(4x - 9) = 24x - 54$ , which differs (test  $x = 0$ :  $-54$  vs  $-36$ ). The others expand to  $24x - 36$ :  $12(2x - 3) = 24x - 36$  and  $4(6x - 9) = 24x - 36$ .
- 20. (B)** Sum  $5 + 8 + 11 = 24$ . Common difference 3 gives  $3n + c$ ; at  $n = 1$  the value 5 means  $c = 2$ , so  $3n + 2$ . Check  $n = 2$ :  $6 + 2 = 8$  correct. Rule  $3n$  gives 3 at  $n = 1$ , wrong.
- 21. (C)** The coefficient of  $x$  is the signed number multiplying  $x$ , here  $-5$ ; the constant term is the one with no letter, 7. Dropping the negative sign gives 5; reading the  $x^3$  coefficient gives  $-1$ .
- 22. (A)**  $n^2 = 16$ , so  $-2n^2 = -32$ ;  $3n = 3(-4) = -12$ . Total  $100 - 32 - 12 = 56$ . Using  $3n = +12$  gives 80; treating  $(-4)^2$  as  $-16$  gives  $100 + 32 - 12 = 120$ .
- 23. (A)**  $x^2$ -terms  $3x^2 - x^2 = 2x^2$  (the count drops, the power stays);  $x$ -terms  $5x + 2x = 7x$ ; constants  $-2 - 7 = -9$ . Result  $2x^2 + 7x - 9$ . Adding exponents to get  $x^4$  is the classic trap.
- 24. (C)** The 60 covers the first 2 km; only  $(d - 2)$  extra kilometres are charged at 15:  $15(d - 2) + 60$ . Test  $d = 2$ :  $15(0) + 60 = 60$  correct. Charging all  $d$  km gives  $15d + 60$ .
- 25. (B)**  $5(2a - 3) = 10a - 15$ ;  $-2(a + 4) = -2a - 8$ . Sum  $(10a - 2a) + (-15 - 8) = 8a - 23$ . Forgetting to distribute the 2 over the  $+4$  (treating it as  $-0$ ) lands on  $8a - 15$ -style traps.
- 26. (C)** An  $n$ -by- $n$  square has  $n^2$  tiles; removing 4 corners gives  $n^2 - 4$ . For  $n = 5$ :  $25 - 4 = 21$ . Forgetting to remove corners gives 25; using  $4n - 4 = 16$  confuses this with a border count.
- 27. (B)**  $2a + b = 2 + 4 = 6$ , so  $(2a + b)^2 = 36$ ; dividing by  $b = 4$  gives  $36 \div 4 = 9$ . Reading  $2a$  as 'a then double' or squaring only part of the bracket changes the numerator; forgetting to square gives  $6 \div 4$ -style errors.
- 28. (A)**  $4(a - 2b) = 4a - 8b$ ;  $-2(2a - 3b) = -4a + 6b$ . a-terms  $4a - 4a = 0$ ; b-terms  $-8b + 6b + 5b = 3b$ . The a-terms cancel, leaving  $3b$ . Sign slip on  $-3b$  inside gives  $-8b - 6b + 5b$  errors.
- 29. (D)** Common difference 4 gives  $4r + c$ ; at  $r = 1$  the count is 12, so  $4(1) + c = 12$  gives  $c = 8$ , hence  $4r + 8$ . Check  $r = 2$ :  $8 + 8 = 16 = 12 + 4$  correct. Using  $4r + 12$  over-counts.
- 30. (D)** Perimeter at  $n = 10$  is 22; at  $n = 6$  is 14; difference  $22 - 14 = 8$ . Equivalently the rule grows by 2 per square over 4 extra squares:  $2 \times 4 = 8$ . Using just  $(10 - 6)$  gives 4.
- 31. (C)** 5 meals cost 5f; 2 desserts at 30 cost 60. Total 5f + 60. A dessert on all 5 days would give 150, but only 2 days have a dessert.

**32. (D)**  $(x + y)/2 = 4/2 = 2$ ;  $(x - y)/2 = 14/2 = 7$ . Difference  $2 - 7 = -5$ . Notice the expression simplifies to  $y$ , and  $y = -5$ . Misreading  $x - y$  as  $9 - 5 = 4$  derails the middle step.

**33. (A)** Remove brackets with sign care:  $3a + 2b - a + b - 2a - 4b$ . a-terms  $3a - a - 2a = 0$ ; b-terms  $2b + b - 4b = -b$ . Result  $-b$ . Not flipping  $-(a - b)$  leaves stray a-terms.

**34. (B)** Triple:  $3x$ . Subtract 4:  $3x - 4$ . Double the whole result:  $2(3x - 4)$ . Test  $x = 2$ : triple 6, minus 4 = 2, doubled = 4;  $2(6 - 4) = 4$  matches.  $2(3x) - 4$  doubles before subtracting, giving a different value.

**35. (B)** Figure 3 has side  $n + 2 = 5$ ; border tiles =  $4(5) - 4 = 16$ . Forgetting the  $-4$  (which removes the double-counted corners) gives 20; the full square is 25.

**36. (C)**  $3p^2 = 3(4) = 12$ ;  $2pq = 2(-2)(3) = -12$ , so  $-2pq = +12$ ;  $q^2 = 9$ . Total  $12 + 12 + 9 = 33$ . Taking  $-2pq$  as  $-12$  gives  $12 - 12 + 9 = 9$ .

**37. (D)** Inside:  $4 - (x - 1) = 4 - x + 1 = 5 - x$ . So  $2x + 6 - (5 - x) + x = 2x + 6 - 5 + x + x = 4x + 1$ . Mishandling the sign on  $(x - 1)$  is the trap. Test  $x = 0$ :  $6 - 5 = 1$  correct.

**38. (A)**  $3(2x - 1) = 6x - 3$ ;  $-(x - 5) = -x + 5$ . Sum  $(6x - x) + (-3 + 5) = 5x + 2$ . Test  $x = 1$ :  $3(1) - (-4) = 7$  and  $5 + 2 = 7$  match. Not flipping  $-(x - 5)$  gives  $5x - 8$ .

**39. (D)** Terms are 3, 5, 7, 9; sum  $3 + 5 + 7 + 9 = 24$ . Using  $2(4) + 1 = 9$  finds only the 4th term, not the sum;  $4 \times 5 = 20$  mistakes the middle value.

**40. (A)** One bouquet:  $r + 2r + 3 = 3r + 3$  flowers. For  $m$  bouquets:  $m(3r + 3) = 3mr + 3m$ . Forgetting to multiply the ferns by  $m$  gives  $3mr + 3$ ; one bouquet only gives  $3r + 3$ .

**41. (C)**  $3 - x = 2$ , so  $(3 - x)^2 = 4$ ;  $2 \times 4 = 8$ ;  $5 - 8 = -3$ . Reading  $2(3 - x)^2$  as  $(2(3 - x))^2 = 16$  gives  $5 - 16 = -11$ ; ignoring the square gives  $5 - 2(2) = 1$ -style errors.

**42. (C)** Remove brackets:  $6m - 2n - 3m + 5n + m - n$ . m-terms  $6m - 3m + m = 4m$ ; n-terms  $-2n + 5n - n = 2n$ . Result  $4m + 2n$ . Not flipping  $-(3m - 5n)$  gives the  $-8n$  trap.

**43. (A)**  $(1/2)(6x - 8) = 3x - 4$ ;  $(1/3)(9x + 6) = 3x + 2$ . Sum  $(3x + 3x) + (-4 + 2) = 6x - 2$ . Test  $x = 0$ :  $(1/2)(-8) + (1/3)(6) = -4 + 2 = -2$  and  $6(0) - 2 = -2$  match.

**44. (B)** Added dots =  $(n + 1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$ . Check  $n = 3$ :  $16 - 9 = 7 = 2(3) + 1$  correct. These are the odd numbers 3, 5, 7, ...; using  $2n$  gives even numbers.

**45. (A)** The 250 covers the first 50 pages; only  $(t - 50)$  pages cost 4 each:  $4(t - 50) + 250$ . Test  $t = 50$ :  $4(0) + 250 = 250$  correct. Charging all  $t$  pages gives  $4t + 250$ .

**46. (C)**  $ab = (2)(-1) = -2$ ;  $bc = (-1)(3) = -3$ ;  $ca = (3)(2) = 6$ . Total  $-2 - 3 + 6 = 1$ . Mishandling  $bc$  as  $+3$  gives  $-2 + 3 + 6 = 7$ .

- 47. (C)**  $5(2a + 3) = 10a + 15$ ;  $-4(a + 2) = -4a - 8$ ;  $-(3a - 1) = -3a + 1$ . a-terms  $10a - 4a - 3a = 3a$ ; constants  $15 - 8 + 1 = 8$ . Result  $3a + 8$ . Not flipping  $-(3a - 1)$  drops the +1, giving  $3a + 6$ .
- 48. (B)**  $A = 2(9) - 5 = 13$ ;  $B = 15 + 1 = 16$ ;  $C = 27 - 10 = 17$ ;  $D = 12 + 4 = 16$ . The greatest is C at 17. Treating  $2x^2$  as  $(2x)^2 = 36$  would wrongly inflate A.
- 49. (D)** Clay for the dolls:  $k$  kg each times  $(k + 2)$  dolls  $= k(k + 2) = k^2 + 2k$ ; plus the fixed 6 kg waste:  $k^2 + 2k + 6$ . Forgetting to distribute  $k$  over both terms gives  $k^2 + 6$ .
- 50. (B)**  $3a^2 = 3(16) = 48$  (square first, then multiply);  $(3a)^2 = 12^2 = 144$  (multiply first, then square);  $3a + 3 = 12 + 3 = 15$ . Confusing  $3a^2$  with  $(3a)^2$  gives 144 for both.
- 51. (A)**  $(4x - y)$ ;  $-2(x - 3y) = -2x + 6y$ ;  $3(y - x) = 3y - 3x$ . x-terms  $4x - 2x - 3x = -x$ ; y-terms  $-y + 6y + 3y = 8y$ . Result  $-x + 8y$ . Sign slip on  $-2(x - 3y)$  gives the  $-4y$  trap.
- 52. (D)** Test  $4n - 1$ :  $n = 10$  gives  $40 - 1 = 39$ . Confirm  $n = 9$  gives 35 (too small) and  $n = 11$  gives 43 (too big). So the 10th term is 39.
- 53. (C)** Saturday  $= 2(6c + 2d) = 12c + 4d$ . Total  $= (6c + 2d) + (12c + 4d) + (4c - 3d)$ . c-terms  $6 + 12 + 4 = 22c$ ; d-terms  $2 + 4 - 3 = 3d$ . Result  $22c + 3d$ . Forgetting to double Saturday gives  $16c$ .
- 54. (D)** First bracket  $2x - 1 = 3$ ; second bracket  $x + 3 = 5$ ; product  $3 \times 5 = 15$ . Adding the brackets instead of multiplying gives 8; this needs two substitutions then a product.
- 55. (D)** The coefficient of  $x$  is the signed number on  $x$ , here +6; the constant term is the one with no letter,  $-9$ . The coefficient of  $x^2$  is  $-1$  (the sign attaches). Dropping signs gives the distractors.
- 56. (D)**  $P = 16$ ;  $Q = 8 + 8 = 16$ ;  $R = 16 + 1 = 17$ ;  $S = 12 + 6 = 18$ . Only P and Q share the value 16;  $R = 17$  and  $S = 18$  are each different. Treating  $2x + 8$  as 'multiply 2 by  $x$  then by 8' or reading  $x^2$  as  $2x$  would break the match.
- 57. (B)** xy-terms  $3xy + 4xy + xy = 8xy$ ; x-terms  $-2x - 5x - x = -8x$ . Result  $8xy - 8x$ . The xy terms keep the xy (powers do not change); merging xy with x is the trap.
- 58. (D)**  $a^3 = 27$ ;  $b^3 = (-2)^3 = -8$ , so  $-b^3 = +8$ . Total  $27 + 8 = 35$ . Treating  $(-2)^3$  as +8 (forgetting an odd power keeps the negative) gives  $27 - 8 = 19$ .
- 59. (B)** Layers have 1, 3, 5, 7 bricks; total  $1 + 3 + 5 + 7 = 16$ . Using  $2(4) - 1 = 7$  finds only the 4th layer; the sum of the first 4 odd numbers is  $4^2 = 16$ .
- 60. (D)**  $7(a - 1) = 7a - 7$ ;  $-3(2a - 5) = -6a + 15$ ;  $2(a + 1) = 2a + 2$ . a-terms  $7a - 6a + 2a = 3a$ ; constants  $-7 + 15 + 2 = 10$ . So  $3a + 10$ ; at  $a = 0$  the value is the constant 10, not 0.